MIL-C-1173E
11 June 1976
SUPERSEDING
MIL-C-1173D
10 February 1969

MILITARY SPECIFICATION

CART, GENERAL HAULING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a manually propelled, open-box-type body cart.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

FSC 3920

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1363	- Measurement of Wood Moisture Content.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The cart shall consist of a wooden open-box-type body, hardwood handle, steel axle, two steel wheels with steel spokes and steel tires, and two support legs.

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3.2 First article (first-produced cart). The contractor shall furnish one or more carts as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce carts that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Wood. The wood used in the fabrication of the cart shall be any of the following species: Ash (white), beech, birch, hackberry, hickory, maple (hard), oak, pecan, or rock elm.

3.3.1.1 Moisture content. At the time of fabrication at the manufacturer's plant, the moisture content of the lumber for the wood boards shall be not less than 8 percent and not more than 12 percent.

3.3.1.2 Quality. The boards shall be free from decay, loose knots (or holes), larvae channels, shake, split, warp, brashness, and slope of grain exceeding 1 inch in 10 inches. Sound tight knots that do not exceed one-third the width of the board in which they appear, and stain which does not obscure the grain will be permitted.

3.4 Design and construction. The cart shall be of the two-wheel type and shall consist of a steel wheel and axle assembly attached to the underside of a wooden open-box-type body by means of wheel hangers and brackets. The body shall be set between the wheels. The wheel-and-axle assembly shall be assembled off-center not less than 2 inches nor more than 2-1/2 inches toward the rear loading end. Support legs shall be affixed to the underside of the body near each end of the body on or in close proximity to the longitudinal centerline; the fixed support leg (at the front handle end of the cart) shall hold the body approximately horizontal when the cart is parked on a level surface, and the folding or adjustable support leg (at the end opposite the handle end, or rear loading end of the cart) shall keep the body from tilting excessively during loading and unloading operations. The body shall have extensions to which a handle assembly shall be attached for pushing the cart. The horizontal centerline of the handle (round wood dowel) shall be not less than 29 inches nor more than 32 inches above ground surface. The wheel and axle assembly shall be designed to keep the wheels from wobbling or moving laterally along the axle during normal operation of the cart.

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3.4.1 Capacity and size. The rated capacity of the cart shall be not less than 500 pounds. The inside body dimensions of the cart shall be not less than 52-1/8 inches long, 30-3/4 inches wide, and 12-1/2 inches deep. The overall cart dimensions shall be not more than 83 inches long x 48 inches wide x 40 inches high.

3.4.2 Body. The body shall be fabricated from boards of uniform thickness and dressed on both surfaces except for the underside of the bottom boards. The body shall be constructed to form a rigid unit when assembled, and shall consist of a floor, two vertical side panels, and two vertical end panels. Joint spacings between board edges shall not exceed 1/16 inch. The end panel at the loading end shall be removable but will engage with the side panels for bracing the side panels in either direction (inward and outward). The end panel at the handle end shall be fixed to the two side panels. The side and end panels shall be reinforced at the corners with steel angles or hardwood cleats, and shall be braced with steel tie rods along the width of the body at the fixed end panel corners. Side panels shall be reinforced with hardwood cleats centrally located the full depth of the side panels (outside surface).

3.4.3 Handle assembly. The handle assembly shall consist of a hardwood dowel (handle) in steel or iron fittings that are attached to hardwood or steel extensions which are connected to the body. The push handle shall be mounted parallel to the wheel axle. Extensions for the handle may be integral with the body or may be rigidly fastened to the body. The dowel shall be round with a diameter not less than 1-3/8 inches nor more than 1-3/4 inches.

3.4.4 Axle hangers and brackets. Axle hangers and brackets shall be fabricated from a ferrous metal material. Hangers and brackets may be used in conjunction with braces and hardwood bolsters.

3.4.5 Axle. The axle shall be steel of one-piece construction. It shall be of sufficient length to support two wheels (one at each end of axle) that straddle the body (widthwise) with sufficient clearance for operation of the cart.

3.4.6 Wheels. Unless otherwise specified (see 6.2), the wheels shall be of nominal 36-inch diameter with steel spokes and steel tires. Wheels shall have grease-sealed bearings, and the wheel hubs shall be provided with lubrication fittings conforming to SAE J534.

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3.4.7 Support legs. Support legs shall be hardwood or metal. Wooden legs shall have a metal ferrule on the end touching the ground surface.

3.4.8 Wheel locking device or brake. A wheel locking device shall be provided for both wheels. The wheel locking device shall be manually operated and shall be an integral part of the cart.

3.4.9 Fastening devices. Fastening devices such as bolts, nuts, cap-screws, washers, locknuts, cotter pins, and similar parts shall be steel. Threads for threaded fasteners shall be in accordance with NBS Handbook H28.

3.5 Performance.

3.5.1 Parking and loading position (no-load). With the cart in park position, the vertical distance from the ground surface to the four corner points on the underside of the body shall not vary by more than 1 inch. In addition, with folding or adjustable leg in loading position, only the fixed support leg and wheels shall contact the ground surface. The clearance between the ground surface and folding or adjustable leg shall be not more than 2 inches.

3.5.2 Static load. The cart shall be capable of bearing a 750-pound static load without evidence of permanent deformation, splits, or any failure of the cart when tested as specified in 4.6.2.3.

3.5.3 Dynamic load. The cart shall be capable of traversing a figure-8 path carrying a load of 500 pounds at a speed of not less than 3 feet per second without evidence of permanent deformation, wobbling of wheels, lateral movement of wheels along the axle, split or cracked boards, or loosening of parts.

3.5.4 Wheel locking device or brake. The wheel locking device or brake, when in brake position, shall be capable of holding the cart stationary (no wheel rotation) on a 10-percent slope when carrying a load of 500 pounds.

3.6 Treatment and painting. Wood and steel surfaces (but not including wheel bearings) shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.7 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130.

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3.8 Workmanship. The cart shall be clean and properly assembled. Burrs, splinters, and sharp edges shall be removed. No parts shall be impaired or damaged in any way, and parts shall show no evidence of rust or corrosion.

3.8.1 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately formed or drilled, and shall have burrs removed. Rivets shall be in accordance with SAE J492 and shall be driven with power tools. The formed heads shall be full, neatly made, concentric with rivet holes, and in full contact with the surface of the member. Excessive upsetting of the rivets to fill holes will not be acceptable.

3.8.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the cart to conform to the examination and test requirements specified in Section 4. Unless otherwise specified (see 6.2), the welding process used in fabrication of the cart shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Individual component inspection (see 4.3).
(b) First-produced cart inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of packaging (see 4.7).

4.3 Individual component inspection.

4.3.1 Sampling. Sampling shall be in accordance with MIL-STD-105. A sample unit shall be one piece of lumber.

4.3.2 Test. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.6.2.1. AQL shall be 2.5 percent defective.

4.4 First-produced cart inspection.

4.4.1 Examination. The first-produced cart shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first-produced cart shall be tested as specified in 4.6.2.2 through 4.6.2.5. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the defects specified in 4.6.1. AQL shall be 2.5 percent defective.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.2 through 4.6.2.5. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The cart shall be examined as specified herein for the following defects:

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Major

101. Dimensions not as specified.
102. Components missing or not as specified.
103. Cart improperly assembled or incomplete, loose, skewed, or tilted.
104. Material bent, bowed, buckled, warped, split, cracked, or not as specified.
105. Board surfaces not dressed where specified.
106. Lubrication or lubricating devices not as specified.
107. Treatment and painting not as specified.
108. Identification marking missing, incomplete, or not legible.
109. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Wood moisture content. The lumber shall be tested for moisture content in accordance with MIL-STD-1363, electric moisture meter method. Three determinations shall be taken on each sample unit, and the result shall be the average of the three determinations. Nonconformance to the requirements of 3.3.1.1 shall constitute failure of this test.

4.6.2.2 Parking and loading position (no-load). Park the cart in its normal position (cart supported by the fixed support leg) on a smooth, level, hard, horizontal surface and position the folding or adjustable leg in its operating position. Measure the vertical distance from the ground surface to the underside of the cart at the four corners and the clearance between the ground surface and the bottom of the folding or adjustable leg. Nonconformance to the requirements of 3.5.1 shall constitute failure of this test.

4.6.2.3 Static load. The cart shall be positioned as specified in 4.6.2.2 with both support legs in operating position and loading end panel removed. A load of 375 pounds shall be equally distributed over an area covering one-half of the cart body load surface (rear half - from loading end to center of the cart body). The load shall be removed after 10 minutes, and the cart body and folding or adjustable support leg shall be inspected. This test shall be repeated with the load equally distributed over the forward half of the cart body load surface, after which the cart body and fixed support leg shall be inspected. Next, a load of 750 pounds shall be equally distributed over the entire loading surface of the cart. The load shall remain on the cart for a period of 30 minutes, after which the load shall be removed and the cart shall be inspected. Nonconformance to the requirements of 3.5.2 under any of the above conditions shall constitute failure of this test.

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4.6.2.4 Dynamic load. The cart shall be positioned on a smooth, level, hard, paved surface. The load surface of the cart body shall have an equally distributed load of 500 pounds. The loading end panel shall be secured in its normal position on the cart body and the folding or adjustable leg shall be in the travel position. The cart shall then be pushed forward manually at the handle to follow a figure-8 path not less than 10 feet and not more than 12 feet (one cycle) at the speed specified for a total of 50 cycles. At the end of the running period, the cart shall be parked, unloaded, and then inspected. Nonconformance to the requirements of 3.5.3 shall constitute failure of this test.

4.6.2.5 Brake. The cart, loaded as specified in 4.6.2.4, shall be parked on a 10-percent slope with the cart's wheel axle parallel to the intersection line formed by the slope and horizontal surface and the handle end of the cart at the higher elevation. The folding or adjustable leg shall be in travel position, and the wheel locking device shall be in brake position. The cart shall rest on the slope for a period of 10 minutes, after which the wheels shall be inspected for evidence of rotation. Nonconformance to the requirements of 3.5.4 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Preproduction pack inspection.

4.7.1.1 Examination. Examine the preproduction pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The preproduction pack of crated components for Level A shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 nor more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of hold-downs, ties, stays, blocking, or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall be considered one defect.
111.	Disassembly not as specified for Level A.
112.	Unpainted metal surfaces not coated with preservative as specified for Level A.
113.	Wheel bearings not lubricated or openings into the wheel hub not sealed as specified for Level A.
114.	Mounting hardware and technical publications not packed together or the box closure not as specified for Level A.
115.	Packing not as specified for Level A.
116.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the first-produced cart or a production cart. When the first article is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first-produced cart and production carts.

5.2 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The wheels, axles, bolsters, if used, wheel hangers and brackets, and support legs shall be disassembled from the cart.

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5.2.1.2 Unprotected surfaces. Unpainted metal surfaces, including the mounting hardware requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Wheel bearings. Each wheel bearing shall be lubricated to capacity with the lubricant recommended by the manufacturer. The opening into the wheel hub shall be covered with barrier material conforming to MIL-B-121, Grade A, Type I, Class 2, and shall be secured in place with tape conforming to PPP-T-60, Type IV.

5.2.1.4 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 and packed in the box with the mounting hardware (see 5.2.1.5).

5.2.1.5 Mounting hardware. The mounting hardware and technical publications (see 5.2.1.4) shall be packed together in a close-fitting box conforming to PPP-B-636, W6c. Box closure shall be in accordance with Method V of the appendix to the box specification.

5.2.2 Commercial. Carts shall be preserved in accordance with FED. STD. No. 356.

5.3 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.3.1 Level A. Each cart, preserved as specified in 5.2, shall be packed in a crate conforming to PPP-C-650, Type II, Style A, except that the bottom crossmember shall be nominal 1- by 6-inch lumber with one edge of the top and bottom crossmember in alinement. Each crate shall be provided with nominal 2- by 4-inch skids laid flat, positioned parallel to the ends, and extending the full exterior width of the crate. Skids shall be placed one under each end crossmember, flush with the inside edge, and under each intermediate crossmember, flush with its unalined edge. The ends of the skids shall be beveled one-half the thickness at a 45 degree angle. Skids shall be secured by nailing through the bottom crossmembers and skids, and the nails clinched not less than 1/8 inch. Nails fastening the skids shall be arranged in two rows in a staggered pattern, with spacing between the nails in each row to be not more than 6 inches. Each row of nails shall be approximately 1/2 inch in from the edge of the skid. The nailing pattern shall begin and end approximately 1-1/2 inches in from the ends of each skid. Strapping shall be positioned alongside the skids. The cart body shall be placed within the crate in

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its normal upright position with the component parts positioned within the crate. Blocking, bracing, and anchoring of contents shall be in accordance with the appendix of the crate specification and in accordance with MIL-STD-1186.

5.3.2 Commercial. Carts, preserved as specified in 5.2, shall be packed in accordance with FED. STD. No. 356.

5.4 Marking.

5.4.1 Level A. Marking for Level A shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The manually propelled cart covered by this specification is intended for general hauling over short distances on prepared and semiprepared road surfaces.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first-produced cart, and number of carts required (see 3.2).
(c) Wheels required, if other than specified (see 3.4.6).
(d) Color required (see 3.6).
(e) When the welding process will not be at the option of the contractor (see 3.8.2).
(f) Time frame required for submission of preproduction pack (see 5.1).
(g) Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First-produced cart. Any changes or deviations of production carts from the approved first-produced cart during production will be subject to the approval of the contracting officer. Approval of the first-produced cart will not relieve the contractor of his obligation to furnish carts conforming to this specification.

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6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the carts in accordance with this specification.

Custodians:
Army - ME
Navy - SA
Air Force - 84

Review activity:
DSA - GS

Preparing activity:
Army - ME

Project No. 3920-0141

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-1173E Cart, General Hauling

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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